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Introduction

Artificial intelligence (AI) is widely recognised as a general purpose technology with the potential to drive significant economic and social changes (Agrawal et al., 2021; Eloundou et al., 2023). Similar to previous 'technological waves', AI may benefit the economy in several ways. By reducing the burden of repetitive tasks and aiding humans to carry out more complex activities, AI is expected to boost productivity, with positive implications for economic growth (Brynjolfsson et al., 2024). AI is also likely to improve the effectiveness of research and development (R&D) activities and foster radical innovation, impacting virtually every country, sector and occupation (Crafts, 2021).

Challenges have also been identified, however. Allowing AI technologies to take autonomous decisions in critical domains such as defence, justice or healthcare raises serious ethical and safety concerns regarding privacy, surveillance, transparency and accountability (Scherer, 2015; Acemoglu, 2021; De Stefano and Taes, 2023)[1]. The use of AI has prompted fears of technological unemployment concentrated among high-skilled occupations previously shielded from automation/digitalisation-related risks. Empirical efforts to test whether these fears are well-grounded and to identify where risks may be most pronounced (e.g. sector, country, occupation) are widespread. Although few occupations seem fully replaceable, a substantial redesign of tasks and workers' skills may be required to take advantage of these technologies (Elondou et al., 2022).

Significant papers introduced 'AI exposure' indicators (Felten et al., 2021, 2023; Webb, 2020) that have now been largely adopted, enabling the first estimations of AI's employment impact (see Montobbio et al., 2023). To date, most empirical studies find no effect or slightly positive employment effects of AI (Albanesi et al., 2023; Damioli et al., 2023; Felten et al., 2021). However, these preliminary findings should be interpreted with caution. Given the early stage of AI development, some relevant features may be neglected by commonly used indicators. More specifically, those indicators tend to provide a static snapshot of current AI developments, only partly capturing the evolving nature of this technology and potentially underestimating its true employment impact.

All studies agree that if benefits such as increased productivity and labour demand are to emerge, these are likely to be asymmetrically distributed, primarily impacting those at the top of the skill distribution, in contrast to previous digitalisation waves. The evidence suggests that positive employment effects are concentrated among high-skilled workers (Felten et al., 2019; Elondou et al., 2023)[2].

[1] Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (European Union (EU) AI Act) came into force on 1 August 2024. It imposes strict regulations on high-risk AI applications, including the use of AI technologies in critical infrastructure such as transport, which may impact the life and health of citizens, as well as in education, vocational training, employment and credit assessments, where they would determine access to essential opportunities and services (https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai).

[2] Such an asymmetric dynamic could be explained by the prevalence of a 'complementarity effect' (implying increasing individual productivity) in the case of AI-exposed high-skill occupations; and of a 'substitution effect' (AI makes an increasing number of medium and low-complexity tasks relatively easier to automate), penalising medium-skilled and low-skilled occupations.

Nevertheless, as Gmyrek et al. (2023)[3] and Elondou et al. (2023) note, the number of tasks in high-skill occupations[4] that generative AI can perform is expected to increase in the near future. Whether this will translate into a stronger complementarity favouring high-skilled workers and giving rise to a further polarisation of labour demand (Felten et al., 2019), or result in substitution effects remains difficult to say.

This skill-related polarisation may interact with other divides, such as geography and gender, exacerbating the asymmetric impact of AI. Gmyrek et al. (2023) estimate that only 0.4% of total employment in low-income countries is at risk of automation, compared to 5.5% in high-income countries. This disparity is largely due to the concentration of AI-exposed occupations in those high-income countries. They also document the gendered impact of AI, with some 3.7% of women's employment worldwide in jobs highly exposed to AI, compared to only 1.4% for men. In high-income countries, these figures double to 7.8% for women and 2.9% for men. This disparity reflects the gender distribution across occupations, particularly the underrepresentation of women in manufacturing and manual jobs, which are relatively less exposed to AI.

Sectoral heterogeneity also needs to be considered. The employment impact of AI has primarily been explored in the context of service-related applications (e.g. generative AI such as Chat-GPT). In manufacturing, for example, AI technologies tend to be 'embodied' in machines and devices (e.g. robots), making their diffusion and impact hard to capture and assess. However, if AI technologies improve the scope and efficiency of automation technologies, their potential impact on employment could be all but negligible.

Different forms of AI can have distinct impacts on the labour market. The general purpose nature of generative AI tools such as Chat-GPT have the potential to enhance workers' productivity in various roles (Brynjolfsson et al., 2024). However, they may also be disruptive in jobs built on writing, content creation and image generation. Using data from the Upwork freelancing platform, Hui et al. (2023) found that ChatGPT had adverse effects on workers in writing-related occupations. By contrast, highly specialised AI systems designed for specific tasks within particular domains (e.g. interpreting medical images in healthcare) are more likely to enhance professional capability and improve diagnostic accuracy (Dembrower et al., 2023), making their impact on the labour market more nuanced. Meanwhile, AI-powered robots have the potential to automate mostly manual tasks, particularly in agriculture, manufacturing and logistics, potentially reducing the need for manual labour (Guarascio et al., 2023).

Heterogeneous effects are likely to emerge even within the same occupations, driven by differences in firm strategies (e.g. product versus process innovation) and varying levels of knowledge and skills among the most exposed workers. Recent evidence suggests that less-experienced or lower-skilled workers may experience the greatest productivity gains from generative AI support (Brynjolfsson et al., 2024; Nurski and Ruer, 2024).

[3] According to Gmyrek et al. (2023), AI differs from previous digital technologies due to 'its refined ability to perform cognitive tasks, such as analysing text, drafting documents and messages, or searching through private repositories and the web for additional information'.
[4] For example, accountants and auditors, clinical data managers, web and digital interface designers.

Overall, a range of structural factors (skills and occupational characteristics, technology, industry, country-level heterogeneities) may shape the AI-employment nexus. Detailed and robust empirical evidence that can explicitly account for the different forms AI can take and the multitude of mediating factors is needed to support the design of relevant policies (e.g. labour, training and social protection policies).

There is a significant gap in the literature on the employment impact of AI in the EU, with much of the available evidence for the United States (US). The first contributions and, significantly, the available AI-exposure indicators are based on the US O*NET[5] repertoire (Felten et al., 2019; Webb, 2020), which provides extensive occupation-level information on skill, tasks and technology use. However, empirical research on Europe is starting to grow (Albanesi et al., 2023), although a number of heterogeneities likely to affect the AI-employment nexus need to be fully investigated, including country and regional differences in sectoral and technological specialisation, labour market institutions, and occupational and skill structure.

This note addresses this gap by mapping AI exposure across the EU Member States and investigating the impact of AI technologies on employment. The analysis builds on Felten et al. (2019, 2021, 2023), adopting an occupation-level perspective. It opens with a systematic review of the recent empirical literature, highlighting relevant differences in measurement approaches and indicators, then presents the data and methods used to assess AI exposure in the EU. It provides a comprehensive set of descriptive evidence at occupation and Member State level, followed by a regression analysis to identify the impact of AI exposure on employment dynamics and test whether specific country-level patterns emerge. Finally, it discusses key findings, highlighting similarities and differences with the existing research and stressing the major policy implications.

Employment impact of AI: literature review

The technology-employment relationship is a frequent topic in economics (Montobbio et al., 2023). From a socioeconomic viewpoint, technology is fundamentally ambivalent: on the one hand, it is a key driver of productivity and economic growth at both micro and macro levels, while on the other, it can be a potential cause of disruptions, particularly in the labour market (i.e. technological unemployment). Such ambivalence is reflected in the contrast between an optimistic, neoclassical approach that assumes the automatic capacity of markets to re-absorb the workers initially expelled due to labour-saving technological change, and a Keynesian or Schumpeterian pessimism that highlights the risks of persistent labour-saving employment effects of technology (see Calvino and Virgillito, 2019). Further developments in the literature emphasise the complex interplay between skills,

[5] The O*NET database contains a rich set of variables that describe work and worker characteristics, including skill requirements.

tasks and technologies (Fernandez-Macias and Hurley, 2016). Should technology have a labour-disruptive effect, it is expected to prevalently concern workers with relatively obsolete competences (the Skill-Biased Technological Change (SBTC) hypothesis[6]) and/or those performing routine, easily substitutable tasks (Routine-Biased Technological Change (RBTC) hypothesis[7]). Structural heterogeneities may also play an important role in determining the ultimate labour impact of technological change: labour market institutions (e.g. relative strength of legal safeguards against socioeconomic risks), degree of unionisation, competitive strategies in firms (e.g. technology and cost-competitiveness strategies), positioning along global value chains (GVCs) and infrastructural/technological elements affecting the diffusion of new technologies (Reljic et al., 2023). Neither indulging in market-based optimism nor the scenario of mass technological unemployment is useful in analysing and understanding the impact of new technologies, including AI.

When examining the employment impact of AI, three changes need to be emphasised (Guarascio et al., 2023). Firstly, AI technologies seem to be capable of accomplishing tasks previously considered unattainable for a machine or digital device, particularly in the service sector (Felten et al., 2023). Activities based on complex reasoning, experience and non-routine cognitive tasks (e.g. coding, software development, interpretation of x-ray images) may soon become domains of AI's 'comparative advantage' (Elondou et al., 2023). As a result, high-skilled and non-routine cognitive tasks could be substituted by AI, potentially exposing some occupations. Secondly, AI is expected to increase the potential of automation technologies, such as robots, increasing the risk of job disruption in manufacturing. Thirdly, structural heterogeneity matters: as knowledge, technology and AI-related capabilities are not homogeneously distributed (Fanti et al., 2022), the rollout of AI technologies may increase market concentration[8] (Kurz, 2023) and polarisation across GVCs. This could see peripheral areas lose further ground, together with the emergence of new forms of techno-economic dependency (Korinek and Stiglitz, 2021). On the positive side, AI may favour the emergence of new tasks and occupations, as well as enhancing workers' productivity on existing activities in services and manufacturing alike (Gmyrek et al., 2023). These productivity gains can translate into lower prices for goods/services on the market, with positive implications for growth and employment ('productivity or price effect', see Calvino and Virgillito, 2019). The introduction of AI may spur product and organisational innovation (e.g. managerial tasks of monitoring, coordinating and directing workers), with

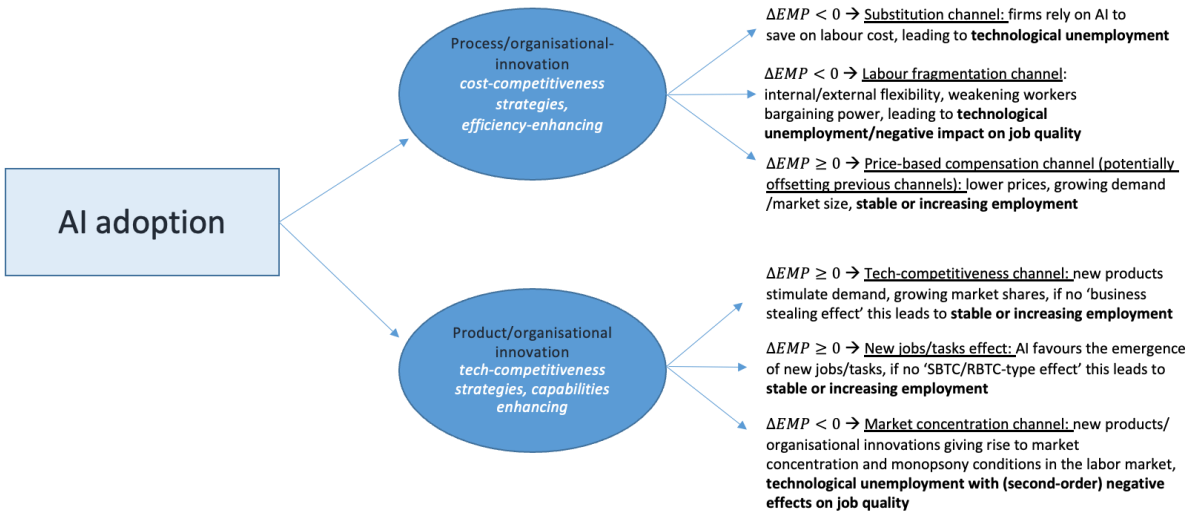
[6] According to the SBTC hypothesis, digital technologies have differentiated effects on the marginal productivity of labour, depending on the level of skills and qualifications of the labour force. New technologies, particularly those related to information and communications technology (ICT), are assumed to be complementary to high-skill jobs (mostly due to the importance of cognitive skills related to the use of computers and ICT devices), while they are expected to penalise medium and low-skilled jobs.

[7] The RBTC approach ranks jobs according to their relative share of routine (i.e. repetitive, encodable, substitutable) tasks rather than their generic skill requirements (see Cirillo et al., 2021).

[8] Although the positive correlation between innovation and market concentration is well-established in economics, digital technologies, particularly those based on the accumulation of large amounts of data may strengthen the position of incumbents, magnifying their capacity to exploit increasing returns (i.e. supply, demand-side economies of scale and network effects), as well as to capture knowledge flows and promising innovation (see Rikap, 2023).

potentially positive effects on labour demand and non-negligible implications for changing occupational and task composition (Chowdhury et al., 2023). Figure 1 presents an illustration of the employment impact of AI, highlighting the various theoretical channels that may be at work.

Figure 1 Employment impact of AI – main theoretical channels



Source: Authors' elaboration.

There are two major channels. Firstly, AI may be adopted to increase efficiency and potentially reduce costs. Typical examples are AI technologies to automate after-sales services, archive management, or logistics tools. In this case, the employment impact can be negative (technological unemployment) if the 'substitution effect' prevails: AI and/or AI-empowered automation technologies would allow production of the same (or a larger) quantity of output, reducing the amount of labour input employed (Vivarelli, 1995; Calvino and Virgillito, 2019). When AI technologies are used to increase organisational flexibility, both internal and external, this may lead to labour fragmentation (e.g. so-called 'algorithmic management'[9]), weakening workers' bargaining power (Cetrulo et al., 2022). A negative outcome can be observed for both labour quality (e.g. increased work pace, contractual precariousness, wage compression) and quantity. On the other hand, if AI-induced efficiency gains translate into a reduction in prices, paralleled by an expansion of markets (and thus labour demand), this can more than compensate for the initial job destruction, leading to a positive employment impact.

[9] According to the growing interdisciplinary literature (see De Stefano and Doellgast, 2023), AI may favour the fragmentation of the workplace by increasing the effectiveness of systems for remote management, monitoring and evaluation of workers' performance, as well as outsourcing entire functions to digital platforms (particularly in the services sector). This may increase the risks of fragmentation/isolation of workers, with negative implications for job quality, degree of unionisation and bargaining power. A typical example is crowd-workers, whose working conditions are extensively described in the literature (Prassl, 2018). In a recent work, De Stefano and Taes (2023) provide qualitative evidence on eight EU Member States on the risks of AI and automated decision-making systems, including applications such as worker surveillance, recording of physical health and mental status, and decision-making on worker hiring, direction, evaluation, and discipline.

A different scenario emerges when AI is associated with the introduction of new products (e.g. AI-empowered digital services, such as generative AI) and/or with capability-enhancing organisational innovation. As new products attract larger demand flows, both production and employment are expected to rise. Similarly, organisational innovations to increase companies' technological capabilities are likely to stimulate the demand for new jobs/tasks (e.g. information technology (IT) professionals, operations specialists, compliance officers)[10]. While this may turn into a positive employment impact, it may disappear if innovators obtain a dominant market share and, ultimately, achieve a monopsony position in the labour market. Under those circumstances, technological unemployment and a deterioration in job quality could occur. These channels are not mutually exclusive: when cost-competitiveness and technological competitiveness strategies are implemented jointly, the ultimate employment impact of technological change, including the introduction of AI, is even harder to predict. Likewise, structural heterogeneities such as the degree of worker protection in the labour market, unionisation, capabilities[11] and infrastructure may facilitate or hinder the adoption of new technologies. Firm size and market concentration factors may also shape the direction and intensity of this relationship.

Given this intricate theoretical background, the empirical literature on the employment impact of AI provides mixed and sometimes discordant evidence. The next section presents a comprehensive review of the literature, by methodology and indicators adopted to capture AI-exposure (mostly), AI-adoption and AI-diffusion (more rarely).

Online job vacancies

The first strategy for assessing the employment impact of AI involves analysing the evolution of labour demand through online job vacancies to understand whether AI-related jobs and skills are becoming a priority in companies' recruitment strategies. Squicciarini and Nachtigall (2021) examined job vacancies in Canada, Singapore, the United Kingdom (UK) and the US, using the Burning Glass Technologies (BGT) data, the largest and most commonly used job vacancy database. AI-related jobs are identified using a set of keywords[12], including more generic terms (such as AI, machine learning), AI applications (e.g. natural language processing (NLP), computer vision), and specific software and libraries (e.g. TensorFlow). Squicciarini and Nachtigall (2021) detected an increasing trend between 2012 and 2018: the total number of AI-related jobs increased across all countries, with a

[10] The adoption of AI technologies for organisational innovation necessitates specialised knowledge and skills to develop, implement and manage AI-driven improvements across various business functions (e.g. IT professionals, operations specialists, compliance officers). For example, a retail company adopting AI for inventory management will need IT specialists to integrate AI with existing inventory databases and supply chain systems, as well as professionals dedicated to maintaining these new systems.

[11] In a broad sense, 'capabilities' refers to a large set of tangible and intangible assets, including industry and firm-level knowledge and technology stock (e.g. R&D, patents), skills and institutions (e.g. universities, public R&D agencies and institutions fostering technology transfer) that may affect 'absorptive capacity' (i.e., a firm's ability to recognize the value of new information, assimilate it, and apply it to commercial ends) at macro, meso and micro-level (Cohen and Levinthal, 1990).

[12] This approach identifies roles focused on AI development as well as those where the use of AI tools is required. AI-related jobs are not confined to specific sectors – although a significant majority are in ICT, they are also found in financial and insurance activities, professional, scientific and technical activities, etc.

peak in the US (almost 150,000 AI-related job postings in 2018). They also reported the growing complexity of such job profiles, in line with the rapid evolution of AI technologies: while skills like software engineering, development and operating systems were predominant in 2012, by 2018, NLP and deep learning had gained prominence, signalling the advent of generative AI.

Acemoglu et al. (2022) used the same database and linked BGT data to SOC (i.e., Standard Occupational Classification System) occupational codes[13] to assess the relative penetration of AI technologies at establishment level in the US. However, they found no clear employment effects of AI at industry or occupation level, likely due to AI technologies relatively limited diffusion. Nevertheless, they documented some changes in occupational composition, with the share of AI-related jobs increasing over time.

Occupational-exposure indicators

A growing number of studies rely on occupational exposure indicators to predict the diffusion of AI technologies and their impact on the labour market. The approach is similar to that of Autor et al. (2003) and subsequent studies (see Montobbio et al., 2023) to measure the degree of routine tasks and related occupational exposure as a proxy of automation/digitisation. Felten et al. (2019) used the O*NET database to define an AI-exposure indicator that measures the 'likelihood that an occupation will come into contact with, be assisted or replaced by AI given the characteristics of the tasks performed and the underlying abilities'. On average, they found that occupations impacted by AI experience no significant change in employment but do see a modest positive change in wages, primarily driven by roles with higher software skill requirements.

Similarly, Brynjolfsson et al. (2018) focused on advancements in machine learning technologies, which are central to almost all AI applications. They introduced a task-based measure of 'suitability for machine learning' (SML),linking it to 18,156 tasks in the O*NET. According to their analysis, a large share of the occupations included in O*NET feature some SML tasks. Few occupations seem to be fully replaceable by AI technologies, although a substantial redesign of their content is required to employ such technologies.

Using a refined version of their previous indicator, Felten et al. (2021) confirmed that occupations most exposed to AI are typically at the top of the skill distribution, such as white-collar jobs. However, their analysis did not explore the prevalence of substituting effects (i.e. higher risk of technological unemployment) or complementary effects (i.e. potentially positive employment impact). To explore this further, Cazzaniga et al. (2024) incorporated additional occupational characteristics into Felten's

[13] Using text-mining, Acemoglu et al. (2022) linked the skill content of job vacancies as reported in the BGT database to job titles (SOC codes), building two measures of 'AI vacancies' – narrow (including a selection of skills relating to AI) and broad (including skills belonging to the broader skill clusters of machine learning, AI, NLP, and data science). These measures are then regressed at establishment-level against the standard O*NET occupation-based AI-exposure indicators (e.g. Felten et al., 2021), showing that the latter perform quite well in predicting the relative intensity of AI-related job vacancies.

indicator to assess whether AI is more likely to complement or replace human labour. These characteristics included factors such as the importance of communication, responsibility and routine task intensity. Their findings indicate that workers at the top of the earnings distribution are more likely to be in occupations with high AI exposure that also have significant potential for complementarity. This could also contribute to increasing income inequality.

Albanesi et al. (2023) and Guarascio et al. (2023) attempted to apply Felten et al.'s (2021) AI-exposure indicator to the European context. Albanesi et al. found a positive correlation between AI exposure and employment at sectoral level, particularly benefiting young and high-skilled workers, while Guarascio et al. observed similar positive employment effects at regional level.

More recently, Felten et al. (2023) updated their AI-exposure indicator to capture the specific impact of advancements in large language modelling (LLM) (the technology behind generative AI such as OpenAI's GPT-4) on employment. Their study identified the occupations most exposed to LLM in the US, including telemarketers and various post-secondary teachers, particularly those specialising in English, foreign languages, literature and history. Legal services and securities, commodities and investments have the highest levels of AI/LLM industry exposure. They also reported a positive correlation between wages and exposure to AI/LLM-related technologies[14].

Using generative pre-trained transformers (GPTs) to measure AI exposure

AI technologies can measure the degree of AI exposure. GPTs can be used to directly identify the most exposed tasks, abilities and occupations, as well as to validate the ranking produced by non-AI-based methods. Gmyrek et al. (2023) used OpenAI's Chat GPT-4 to estimate task-level scores of occupational exposure to generative AI and then examine heterogeneities across countries around the world. Their research indicated that the most significant impact – whether augmenting or automating – is expected in high and upper-middle income countries, attributable to a greater percentage of employment in highly exposed occupations. Consistent with earlier research, they observed that generative AI exposure is primarily concentrated in clerical occupations, with 24% of clerical tasks highly exposed. Elondou et al. (2023) adopted a mixed method, combining expert opinion and GPT-4 classifications to assess the impact of AI on the US labour market. Focusing on the 2020-2021 period, they showed that around 80% of the US workforce might see at least 10% of their tasks affected by AI/LLM-related technologies, with nearly one-fifth of occupations potentially having up to 50% of their tasks impacted.

AI and patent-based indicators

Another way to measure AI exposure that considers both technological and occupational heterogeneities is to track the co-occurrence of verb-noun pairs in patent titles and task descriptions (e.g. in the O*NET database). Using this approach and focusing on the US, Webb (2020) showed that AI is

[14] In line with their previous contributions (Felten et al., 2019, 2021), Felten et al. (2023) remained neutral on AI-related complementarity/substitution effects, i.e. they made no assumption in building their AI-exposure indicator, nor did they push the interpretation of their results towards the prevalence of either effect.

more likely to affect skilled and older workers compared to previous technological waves (e.g. robots or software). While this reinforced earlier evidence, the robustness of their patent-occupation indicator is partly undermined because patent titles do not contain a full description of the underlying technology. In addition, limiting the analysis to verb-noun pairs may have increased the likelihood of false positives (Guarascio et al., 2023).

Firm-level studies

Recent studies have shifted from examining AI exposure to focusing on the actual adoption of generative AI and its impact on employment. Brynjolfsson et al. (2024) analysed data from 5,000 customer support agents and found that generative AI boosts productivity, particularly among novice and low-skilled workers, while improving employee retention. Similarly, Hui et al. (2023) explored the effects of ChatGPT's release on online freelancers. In this case, they reported a 2% decrease in demand for freelancers on the Upwork platform, along with a 5.2% decline in monthly earnings for those in writing-related jobs, compared to freelancers in less-affected occupations. These findings should be interpreted with caution, as the scope of the studies is relatively narrow and their results are not broadly generalisable. They reflect specific occupations (e.g. customer service agents) and particular contexts (online platforms), rather than providing insight into the broader labour market impacts of generative AI.

Due to the scarcity of firm-level data on AI adoption, few studies rely on proxies for innovation, such as actual patent counts. Damioli et al. (2023) analysed a sample of 3,500 AI innovators over the period 2000-2016 and found a positive short-term employment impact of AI patenting of around 3-4%, along with an analogously positive effect of other (non-AI) firm innovation activities. Similarly, Yang (2022) examined the Taiwanese electronic industry and employed a keyword matching technique to identify AI innovators, with the results confirming the positive employment impact of both AI and non-AI patents. However, by focusing exclusively on patenting companies, these studies do not address the net aggregate effect of AI adoption.

Table 1: Overview of selected studies on the impact of AI on employment

Authors, publication year	**Methods, sample**	**Main results**
Online vacancies		
Squicciarini and Nachtigall (2021)	BGT data; Canada, Singapore, UK and US; 2012-2018	Growing demand for AI-related jobs and skills
Acemoglu et al. (2022)	BGT linked with US Bureau of Labour Statistics data; US; 2010-2018;	No significant effects of AI at occupation and industry level
Occupational exposure indicators		
Brynjolfsson et al. (2018)	O*NET database; US	Most occupations include at least some at-risk-of-substitution tasks; few occupations fully substitutable by AI

Felten et al. (2019, 2021)	O*NET database; US; 2010-2016	Positive association with employment growth
Albanesi et al. (2023)	O*NET/ISCO database; 16 EU Member States (industry-level); 2011-2019	Positive association with employment growth, mostly high-skilled workers
Guarascio et al. (2023)	O*NET/International Standard Classification of Occupations (ISCO) database; EU regions; 2011-2019	Positive association with employment growth
Cazzaniga et al. (2024)	O*NET database; global	Workers at the top of the earnings distribution more likely to be in high AI-exposed occupations that also have significant potential for complementarity
Gmyrek et al. (2023)	ChatGPT-based indicators; global	Generative AI mostly impacts clerical occupations (24% of tasks highly exposed)
Elondou et al. (2023)	Mixed method: ChatGPT and expert opinion; US; 2020-2021	Almost one-fifth of occupations could have up to 50% of their tasks impacted
AI patents and occupation-based indicators		
Webb (2020)	O*NET/Google patents; US; 2000-2018	AI more likely to affect skilled and older workers
Firm-level and case studies		
Damioli et al. (2023)	Orbis-Bureau Van Dijk linked with patent data; top AI innovators; 2000-2016	Positive short-term employment impact of AI patenting
Yang (2022)	Taiwan; top AI innovators in electronics industry; 2002-2018	Positive employment impact of both AI and non-AI patents
Hui et al. (2023)	Upwork freelancing platform	Freelancers experience 2% decrease in the number of monthly jobs after the introduction of generative AI
Brynjolfsson et al. (2024)	5,179 customer support agents	Access to generative AI increases the productivity and retention of individual workers, with novice and low-skilled workers benefiting most

Source: Authors' elaboration.

Data and methods

DATA DESCRIPTION

To investigate the employment impact of AI in the EU, data from various sources[15] have been used to construct a country-occupation dataset covering the 2012-2022 period. This timeframe was selected for two key reasons: the major revision of the ISCO classification in 2011 and the relatively recent diffusion of AI technologies, which makes earlier periods less relevant.

AI exposure is based on the AI exposure indicator proposed by Felten et al. (2021), which links 10 AI applications to 52 distinct workplace abilities listed in the O*NET[16]: abstract strategy games; real-time video games; image recognition; visual question answering; image generation; reading comprehension; language modelling; translation, speech recognition; and instrumental track recognition. This selection is informed by the Electronic Frontier Foundation, which identifies the AI domains with the most scientific progress and the most promising applications expected in the short to medium term. AI applications were matched to workplace abilities based on a survey administered to 2,000 individuals via Amazon's Mechanical Turk (mTurk) platform, where interviewees were asked to assess the relevance of each AI application to 52 abilities (see Felten et al., 2021).

Occupational AI exposure (AIOE) is calculated by weighting the AI exposure (A_j) of these abilities (j) by their prevalence (L_{jk}) and importance (I_{jk}) in each occupation (k)[17]:

$$AIOE_k = \frac{\sum_{j=1}^{52} A_j * L_{jk} * I_{jk}}{\sum_{j=1}^{52} L_{jk} * I_{jk}} \quad (1)$$

Assuming that AI-related workplace abilities in the US are similar to those in the EU (Albanesi et al., 2023; Guarascio et al., 2023), 774 six-digit SOC are reclassified to 427 four-digit ISCO-08 occupations[18]. This classification is further aggregated by calculating the mean exposure across 126 three-digit ISCO-08 occupations. Notably, this methodology remains neutral on whether AI substitutes or complements occupations (Felten et al., 2021), i.e. higher AI exposure does not imply a higher risk of substitution by AI technologies.

[15] US O*NET repertoire, the European Labour Force Survey (EU-LFS), Eurostat and OECD Inter-country Input Output (ICIO) tables.

[16] Oral comprehension, speech recognition, speech clarity, stamina, originality, memorisation, manual dexterity, among others.

[17] For each occupation, ability-level AI exposure as calculated by Felten et al. (2021, p. 2201) is multiplied by the prevalence and importance scores reported in the O*NET database. This considers the relative prevalence/importance of each ability for each occupation: 'abilities that are integral to an occupation have high prevalence and importance scores, while those that are used less often or are less vital have lower prevalence and importance scores. For example, for CEOs, O*NET considers abilities such as oral comprehension (importance: 4.5; prevalence: 4.88) and oral expression (importance: 4.38; prevalence: 5) as highly important, and abilities such as speed of limb movement and static strength as nonessential (for both, importance: 1; prevalence: 0)'.

[18] The use of O*NET-based indicators to assess the evolution of the European labour market implies translating occupation-level variables from the SOC to ISCO classification and assuming that occupational characteristics are quite homogeneous in the US and Europe. Although not trivial, this assumption is reasonable, with influential works using O*NET indicators to study RBTC in Europe (Goos et al., 2009, 2014), given the similar degree of technological development of their production structures. This approach is necessary as information similar to that provided in O*NET is not available for European countries and the most granular occupation-level data available in Europe is at four-digit level.

AI exposure is also calculated for each EU Member State by linking AIOE data with country-occupation employment weights derived from the EU-LFS:

$$AICE_{i,t} = \sum_{k=1}^{126} \frac{EMP_{k,i,t}}{EMP_{i,t}} * AIOE_k \quad (2)$$

where $EMP_{k,i,t}$ denotes the number of employees in occupation k in country i for year t, while EMP_{it} is the total number of employees in country i for year t. The $AIOE_k$ term stands for the AIOE, as defined in Equation 1. The AI country exposure (AICE) indicator is then normalised to a zero mean and unit standard deviation to represent the relative AI exposure levels across countries[19].

These two indicators map AI exposure patterns across Europe, reflecting country-specific occupational and sectoral specialisation, as well as investigating the relationship between AI technologies and employment growth.

Workforce characteristics. Data from the EU-LFS are used to construct a set of socio-demographic variables. These include the share of women, the distribution of contract types (share of permanent vs temporary contracts), age groups (20-34, 35-54, 55-64), education level (lower secondary, upper secondary, tertiary education) and country of birth (foreign vs native) within each ISCO three-digit occupation.

Additional variables. The regression analysis includes a set of country-level controls that may affect the AI-employment nexus as a robustness check instead of country fixed effects (FE). To account for each country's position within GVCs, a ratio between forward and backward participation[20] was calculated using the OECD's ICIO tables (Baldwin and Freeman, 2022). More specifically, a value higher than 1 indicates net gains from participation in GVCs, while a value lower than 1 indicates net losses. In addition, data on public R&D are sourced from Eurostat, while information on public procurement comes from the OECD database.

METHODS

The impact of AI on the EU labour market is investigated by adopting a multi-dimensional approach that encompasses descriptive and econometric analyses.

Firstly, AI exposure across EU occupations and countries is mapped using the indicators described in Equations 1 and 2. This mapping enables the identification of occupations that are more exposed to AI technologies and the relative positions of EU Member States in AI exposure.

[19] Countries displaying higher (and positive) values are considered strongly exposed, while those displaying lower (and negative) values are less exposed.
[20] Backward participation refers to the share of foreign added value activated by a country's exports (use of raw materials, intermediate inputs and services provided by other countries in prior segments of the GVC), while forward participation refers to the share of a country's added value activated by other countries' exports.

Secondly, as this indicator is neutral on whether AI substitutes or complements labour, it is examined considering the routine task intensity of occupations, in line with the RBTC literature (Acemoglu and Autor, 2011; Goos et al., 2009)[21]. This enables assessment of whether or not the impact of AI technologies is 'mediated' by task routineness. Focusing on highly AI-exposed occupations, the study tests whether occupations characterised by a high share of routine tasks face greater substitution risks than other occupations (Brynjolfsson et al., 2024). It also examines whether and the extent to which non-routine occupations are complementary to these new technologies, as in previous digitalisation waves (Cirillo et al., 2022), or open to substitution risks due to the peculiar characteristics of AI.

The theoretical rationale is as follows: occupations with predominantly repetitive and easily codifiable tasks have historically been substituted by technology to a greater degree (Autor et al., 2003). Despite significant differences between AI and earlier waves of digitalisation, routine-intensive occupations are likely to remain more exposed to substitution or, at the very least, are less likely to benefit from AI technologies. For instance, while both numerical clerks and university and higher education teachers have high levels of AI exposure, their tasks differ significantly. Numerical clerks perform mainly routine cognitive tasks such as data entry and processing documents, which are easily handled by AI systems without human supervision. By contrast, university and higher education teachers require human judgment and interaction, making full automation more challenging. However, the remarkable progress of AI-based systems in performing complex tasks (e.g. interpreting images and sound, coding, content creation, reading/interpreting legal and medical records) may threaten even non-routine-based occupations, such as radiologists, ICT specialists and legal assistants. The early stage of the most promising AI applications is accompanied by non-trivial limitations, such as the need for massive computing power and training systems, as well as the constraints imposed by recent regulations on privacy and ethical concerns (e.g. the EU AI act). These limitations may *de facto* shield occupations from potentially unfavourable labour market outcomes (Cazzaniga et al., 2024).

Operationally, the study relies on Lewandowski et al.'s (2020) occupational taxonomy, which classifies each ISCO three-digit occupation into one of five task groups based on the predominant task type: non-routine cognitive analytical (e.g. mathematicians, actuaries and statisticians), non-routine cognitive personal (e.g. secondary education teachers), routine cognitive (e.g. general office clerks), non-routine manual (e.g. domestic, hotel and office cleaners and helpers) and routine manual (e.g. manufacturing labourers).

Thirdly, it explores potential disparities among workers in highly AI-exposed occupations, focusing on sociodemographic characteristics such as skills, age, gender, ethnicity and contract types. This analysis aims to provide policymakers with insights on how to prevent and mitigate the adverse effects of AI.

Fourthly, an econometric analysis assesses the impact of AI exposure on employment growth at country-occupation level[22] by estimating the following model specification:

[21] This exercise is exploratory in nature and further investigation based on more granular information about AI applications and occupational characteristics will be necessary to properly map substitution risks and complementarities.
[22] Analysis includes all EU Member States except Bulgaria, Malta and Slovenia, which are missing information on socio-demographic characteristics at three-digit occupational level.

$$\Delta Y_{i,j} = \alpha_0 + \beta_1 AIOE_j + \gamma * x'_{i,j} + \tau_i + \varepsilon_{ij} \quad (3)$$

where the dependent variable ($\Delta Y_{i,j}$) is the annual employment growth rate over the period 2012-2022 in country *i* and occupation *j*; AIOE denotes occupation-level AI exposure. The vector x' comprises a set of control variables at country-occupational level referring to 2012 (initial year)[23], τ_j denotes country FEs, and ε is the error term. The coefficient of interest is β_1, which gives the sign and magnitude of the relationship between AI exposure and employment growth, all else being equal. More specifically, it measures how much an increase of one standard deviation in the AIOI score is associated with an increase in employment growth.

To further investigate whether the impact of AI varies across different occupations, particularly those characterised by routine cognitive tasks, the model in Equation 3 is augmented by including an interaction term between AI exposure and a binary variable (routine cognitive tasks, RC), where 1 represents occupations based on routine cognitive tasks and 0 represents other occupations.

$$\Delta Y_{i,j} = \alpha_0 + \beta_1 AIOE_j + \beta_2 (AIOE_j * RC_j) + \beta_3 RC_j + \gamma * x'_{i,j} + \tau_i + \varepsilon_{ij} \quad (4)$$

A significant β_2 would indicate that the degree of routine task intensity mediates the impact of AI on employment.

Equations 3 and 4 are estimated using pooled ordinary least squares (POLS) and FE methods[24]. Endogeneity is not a major concern here, as AIOE exposure was constructed using O*NET data referring to US occupations, which plausibly exogenously influence employment dynamics in EU Member States (Albanesi et al., 2023). In addition, as the AI exposure indicator measures the degree of overlap between AI applications and human abilities, it tends to capture technological progress rather than demand dynamics. Consequently, AI exposure is not directly influenced by employment levels in the way actual AI adoption would be, further mitigating the risk of endogeneity.

Finally, as average estimates based on the pooled sample of countries may mask important cross-country differences, the model in Equation 4 is estimated separately for four groups of EU Member States based on their innovation performance as per the European Innovation Scoreboard (European Commission, 2023): *Innovation Leaders* (Belgium, Denmark, Finland, the Netherlands, Sweden), *Strong Innovators* (Austria, Cyprus, France, Germany, Ireland, Luxembourg), *Moderate Innovators* (Czechia, Estonia, Greece, Hungary, Italy, Lithuania, Portugal, Spain) and *Emerging Innovators* (Croatia, Latvia, Poland, Romania, Slovakia). This classification reflects four distinct national innovation systems, highlighting significant cross-countries differences in human resources, digitalisation, public

[23] Share of native citizens, senior workers and men, as well the share of those with upper secondary education and those with university degrees.
[24] Standard errors are adjusted for heteroscedasticity (non-constant variance of the residuals) and clustered at country level to account for serial correlation.

and private investment and overall innovation capabilities (European Commission, 2023), which could affect the labour-AI nexus.

MAPPING AI EXPOSURE IN THE EU

This section maps the exposure of occupations and countries to AI technologies.

OCCUPATIONS

Table 2 ranks occupations according to their degree of exposure to AI technologies. Occupations with the highest AI exposure levels – predominantly within the professional category (ISCO 2) – tend to require high levels of education alongside complex quantitative and analytical skills. This observation aligns with growing concerns around the potential for technological unemployment among highly skilled workers. However, higher AI exposure could also indicate a complementarity with these technologies, where workers in such occupations, involving mainly non-RC tasks, may become more productive thanks to AI-induced reshuffling of tasks, potentially leading to gains in employment and wages.

Conversely, occupations least exposed to AI are typically manual and demand lower skill levels, notably elementary occupations (ISCO 9). This pattern is consistent across all sectors – agriculture, mining, manufacturing and services – where occupations primarily involving physical tasks remain relatively unexposed to AI[25]. This contrasts with the impact of industrial robots, which predominantly affected manual workers (Fernandez-Macias et al., 2021; Reljic et al., 2023).

Table 2: Top 10 most and least AI-exposed occupations

Ranking	ISCO 3-digit	ISCO 3-digit label	Type of prevalent task
Top 10			
1	212	Mathematicians, actuaries and statisticians	NRCA
2	241	Finance professionals	NRCA
3	261	Legal professionals	NRCA
4	242	Administration professionals	NRCA
5	431	Numerical clerks	RC
6	231	University and higher education teachers	NRCA
7	411	General office clerks	RC
8	122	Sales, marketing and development managers	NRCP
9	251	Software and applications developers and analysts	NRCA
10	233	Secondary Education Teachers	NRCP

[25] While AI technologies may impact manual activities in the near future (e.g. smart robots), as yet their diffusion and labour market impact is limited.

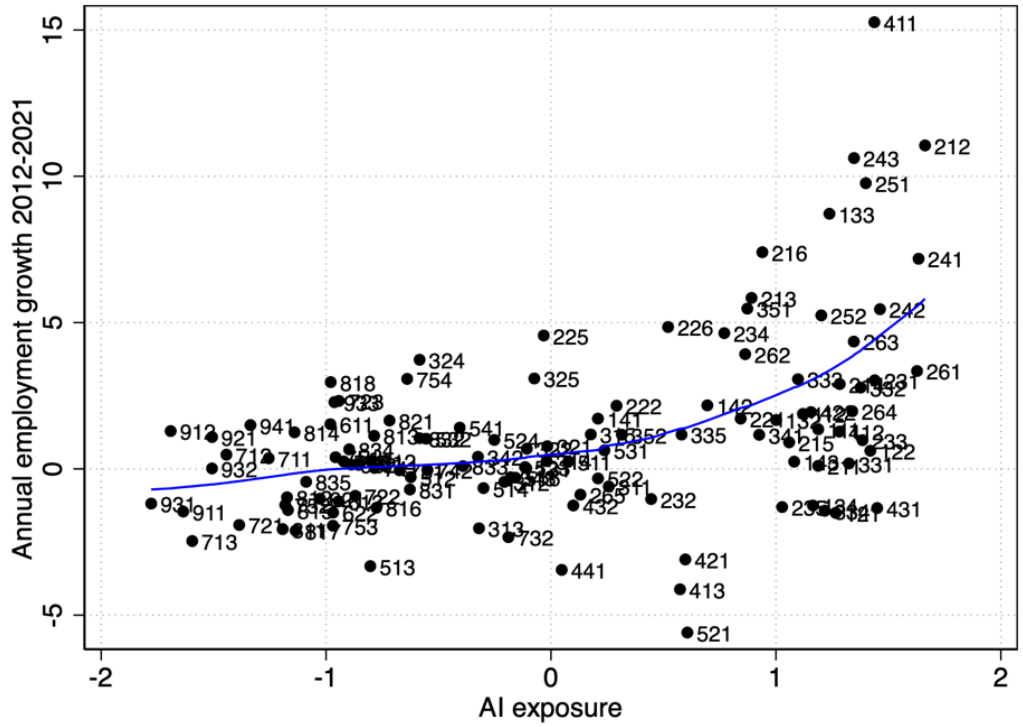
Bottom 10			
117	712	Building finishers and related trades workers	NRM
118	633	Subsistence mixed crop and livestock farmers	NRM
119	921	Agricultural, forestry and fishery labourers	NRM
120	932	Manufacturing labourers	RM
121	634	Subsistence fishers, hunters, trappers and gatherers	NRM
122	713	Painters, building structure cleaners and related trades workers	RM
123	911	Domestic, hotel and office cleaners and helpers	NRM
124	912	Vehicle, window, laundry and other hand cleaning workers	RM
125	631	Subsistence crop farmers	NRM
126	931	Mining and construction labourers	NRM

Notes: NRCA = non-routine cognitive analytical occupations; NRCP = non-routine cognitive personal occupations; RM = routine manual; NRM = non-routine manual.
Source: Authors' elaboration.

These highlight a novel pattern where AI shows the capacity to undertake mostly non-RC tasks that were beyond earlier generations of automation technologies (Guarascio et al., 2023). However, the AI exposure indicator differs from other well-known occupational measures, such as routine task intensity. Although non-RC tasks prevail in most highly exposed occupations, two of the top 10 occupations are numerical and general clerks, whose tasks are predominantly routine. The exposure of manual tasks to AI is uniformly low, regardless of routine intensity levels.

Figure 2 shows the relationship between annual employment change at the three-digit occupation level and AI exposure at three-digit ISCO level within the 27 EU Member States (EU-27) between 2012 and 2021. Despite some noise, a positive correlation emerges, with a correlation coefficient of 0.46. This suggests that occupations more likely to be affected by AI have seen more significant growth rates over the last decade. This is an unconditional correlation and does not imply causality, as other factors may be influencing these dynamics (e.g. heterogeneous macroeconomic conditions at country/sector and thus occupational level).

Figure 2 Employment growth and AI exposure, 2012-2021, EU-27

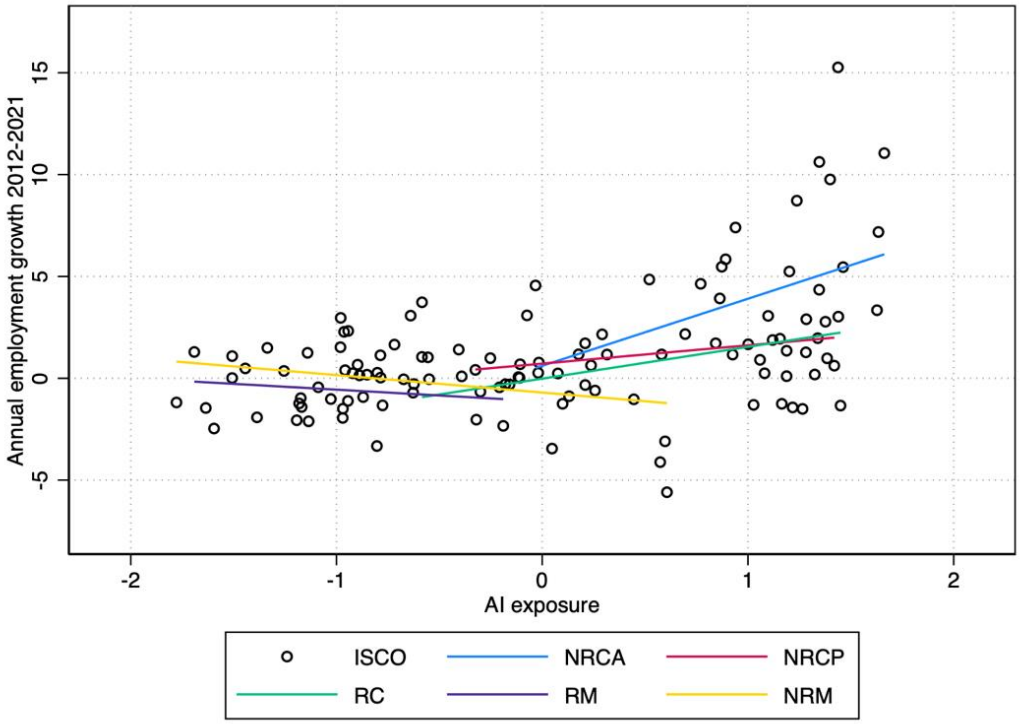


Notes: AI exposure refers to 2021. The annual employment growth rate was calculated for each ISCO three-digit occupation using EU-LFS data for the EU-27 aggregate over 2012-2021. Marker labels refer to ISCO 3-digit codes.
Source: Authors' elaboration.

The pattern and strength of this relationship vary by the type of task associated with the occupation, as shown in Figure 3. The positive correlation is primarily driven by the NRCA group, which has a steeper positive slope, implying that occupations involving predominantly NRCA tasks tend to experience higher employment growth. By contrast, the RM and NRM groups exhibit a relatively flat slope, indicating a negligible relationship between AI exposure and employment growth and suggesting that manual tasks, routine or not, are less susceptible to AI.

Occupations predominantly characterised by RC tasks do not show negative employment change, suggesting that AI is not yet negatively correlated with these professions. However, this does not rule out the possibility that these occupations could become more susceptible to substitution by AI technologies in the near future. Similar considerations apply to manual and non-RC occupations, as the growing number of AI applications may complement/substitute tasks even in these domains.

Figure 3 Employment growth and AI exposure, by type of prevalent task



Notes: NRCA (blue) = non-routine cognitive analytical; NRCP (red) = non-routine cognitive personal; RC (green) = routine cognitive; RM (purple) = routine manual; NRM (yellow) = non-routine manual occupations; Annual employment growth rate calculated using EU-27 data.
Source: Authors' elaboration.

Table 3 presents the occupations most exposed to AI, specifically those within the top quartile of AI exposure[26], using EU aggregate data for 2022 to present their distribution by key sociodemographic characteristics: gender, age, country of birth, contract type and educational level.

Table 3 shows that more than half (54.3%) of workers in top AI-exposed occupations are women. This observation supports earlier findings (Cazzaniga et al., 2024; Gmyrek et al., 2023) that women are more likely to be impacted by AI than men, which can be attributed to gender imbalances across occupations and sectors. Women are less likely to be employed in manufacturing sectors and in roles that are intensive in manual tasks, which are less exposed to AI. It also shows a stark contrast in educational attainment for top AI-exposed occupations compared to the total workforce, reflecting higher skills requirements in these jobs.

These observed heterogeneities highlight the non-neutral nature of technological change, indicating that the impacts of AI are unlikely to be uniformly distributed across different worker groups. This means that tailored policy measures are necessary to ensure that the benefits of technological progress are distributed more equitably.

[26] The quartiles categorise three-digit ISCO occupations into: high, medium-high, medium-low, and low AI exposure.

Table 3: Workforce characteristics, EU-27, 2022

	Total	Top AI exposure
Gender		
Women	46.8%	54.3%
Men	53.2%	45.7%
Age		
20to34	27.9%	27.1%
35to49	38.8%	41.2%
50to64	33.3%	31.8%
Country of birth		
Foreign	14.0%	10.2%
Native	86.0%	89.8%
Contract		
Permanent	87.0%	89.6%
Temporary	13.0%	10.4%
Education		
Lower secondary	15.7%	3.5%
Upper secondary	46.6%	29.8%
Tertiary	37.7%	66.7%

Source: Authors' elaboration based on EU-LFS data.

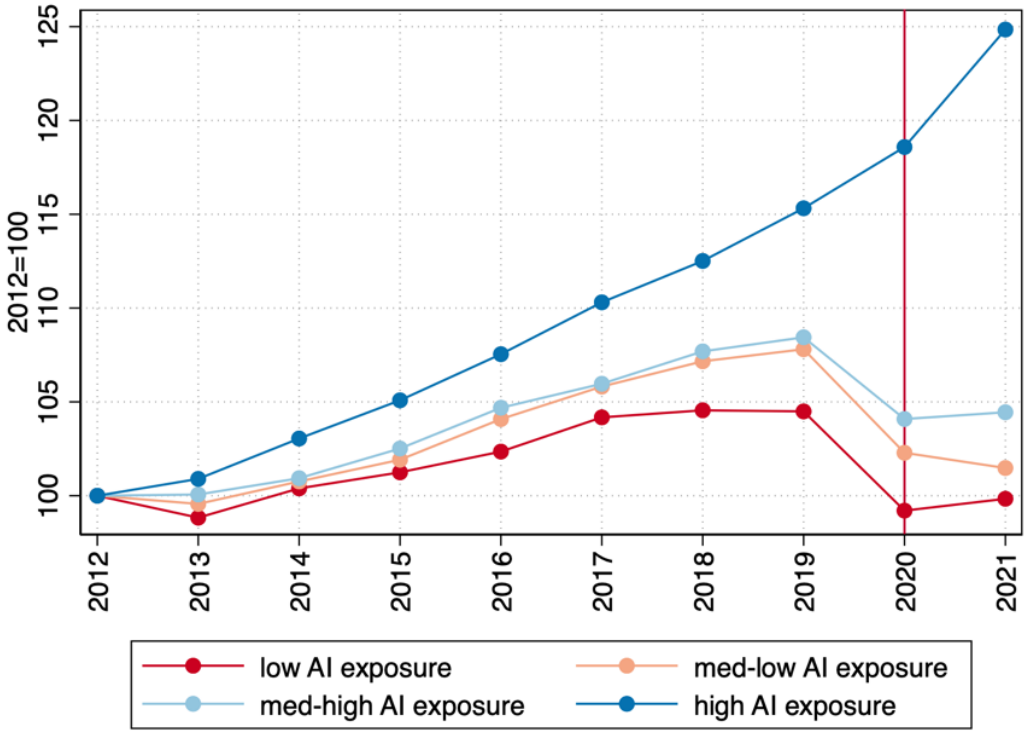
COUNTRIES

Building on the occupation-level analysis, the focus now shifts to the interplay between AI exposure and employment dynamics across the EU Member States, offering insights into the resilience and adaptability of the EU labour market in the face of AI advancements.

Figure 4 presents employment growth in the EU over the last decade, focusing on four occupational groups, differentiated by their degree of AI exposure. The quartiles categorise three-digit ISCO occupations into: high, medium-high, medium-low, and low AI exposure[27].

[27] Thresholds coincide with the quartiles of the AI exposure indicator dividing its distribution into four equal parts.

Figure 4 Employment growth, by AI exposure quartile, 2012-2021, EU-27



Notes: Occupations were split into four quartiles according to the level of AI exposure, ranging from lowest to highest; annual employment growth rate calculated separately for each quartile using EU-27 aggregate data. Source: Authors' elaboration.

From 2012 to 2019, all groups witnessed employment growth, albeit at varying rates. Notably, occupations with low AI exposure showed less pronounced growth (5%), with a stagnation phase between 2017 and 2019. Medium-high and medium-low AI exposure occupations followed a similar trajectory, diverging only with the onset of the COVID-19 pandemic and the recovery phase.

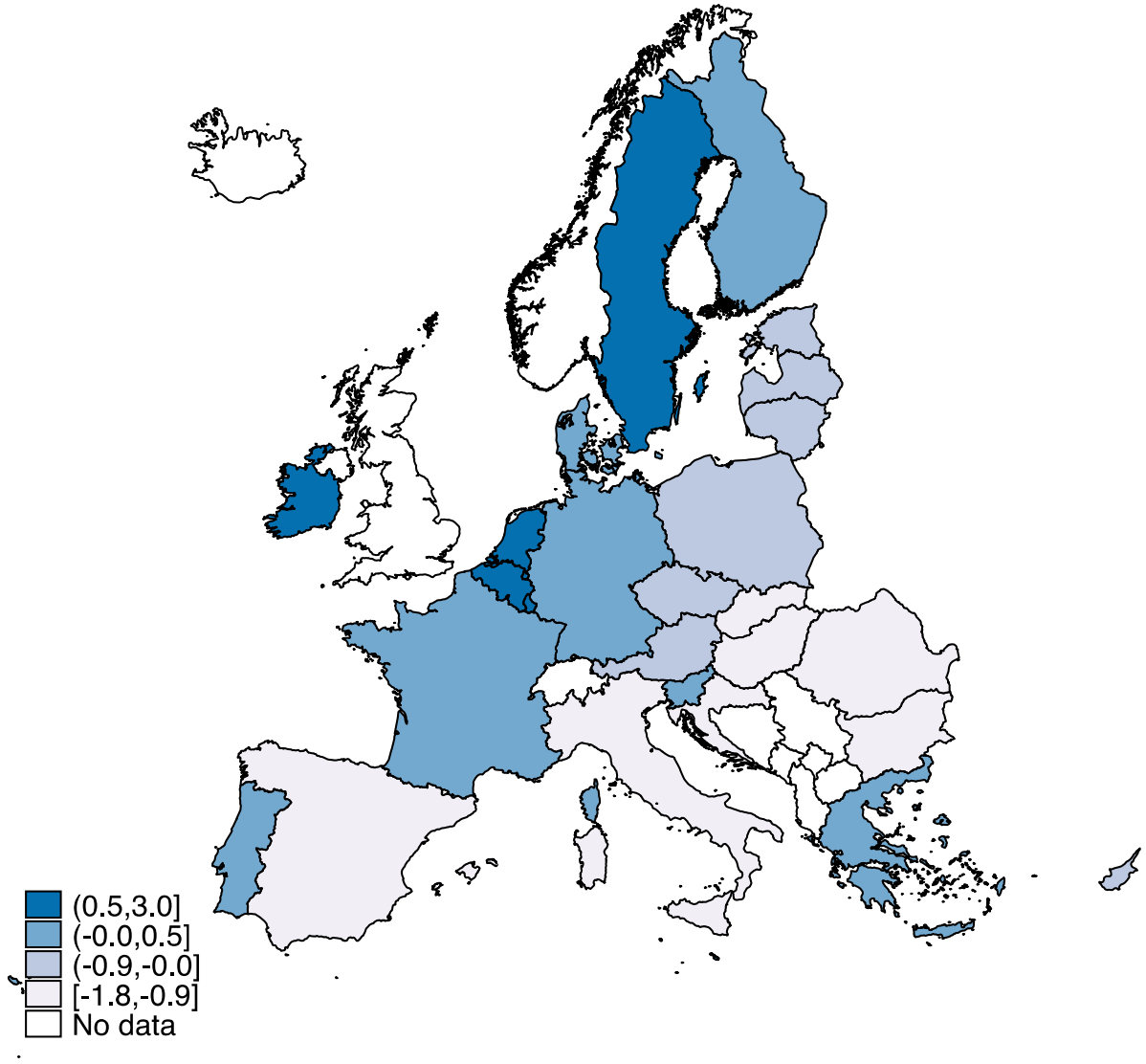
By contrast, occupations highly exposed to AI technologies experienced uninterrupted employment growth, demonstrating resilience even during the COVID-19 pandemic, which significantly disrupted labour markets across the EU. The pandemic-induced shift towards remote work, coupled with accelerated adoption of AI technologies, not only helped to preserve existing jobs but appeared to facilitate the creation of new jobs in technology-complementary roles. This dynamic (at least partly) explains the increase in employment in high AI exposure occupations in 2021. While the other three groups remained well below their pre-crisis trends, high AI exposure occupations saw a year-on-year increase, with a particularly sharp rise from 2019 to 2021[28].

While indicative of a general trend, these aggregate employment dynamics may mask significant heterogeneities among the Member States. Capacity to adopt and then harness the benefits of AI

[28] Of 30 occupations in the high AI exposure group, two-thirds exhibited positive employment growth in 2020. The highest employment growth was observed among: database and network professionals; software and applications developers; analysts; and general office clerks. Conversely, only six occupations reported a decline in employment greater than 5%, with no associated rebound in 2021 for administrative and specialised secretaries, business services and administration managers, and professional services managers.

technologies is likely to differ across countries due to their distinct structural characteristics, possibly complementing labour in certain areas while substituting it in others (Reljic et al., 2023).

Figure 5 AI exposure, 2021, EU-27



Source: Authors' elaboration.

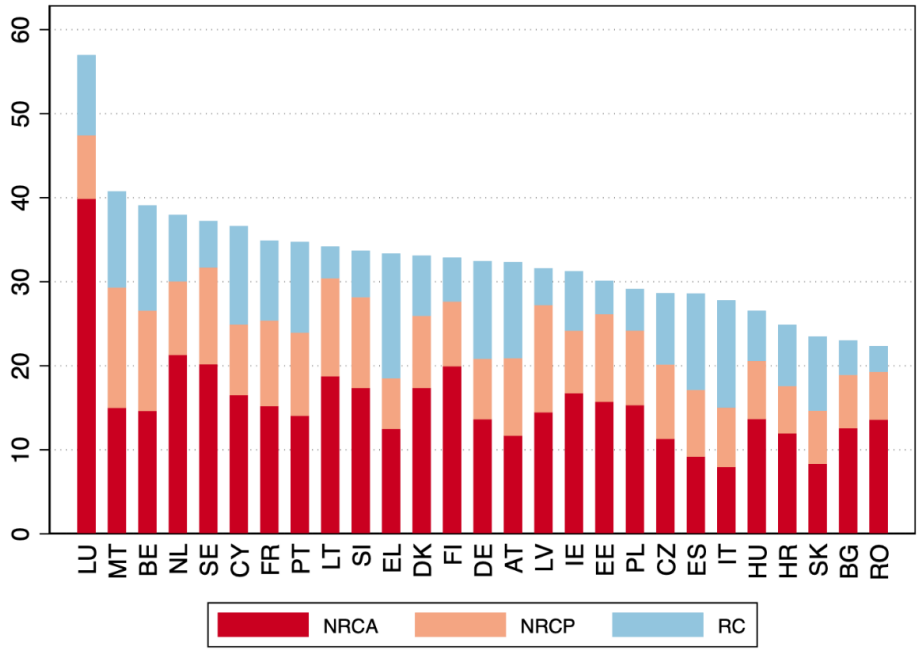
Figure 5 presents the relative AI exposure across EU Member States in 2021 (see Annex 1 for full list). Western and Northern Member States, such as the Scandinavian countries, Luxembourg, Belgium and the Netherlands, exhibit high AI exposure. These are service-based economies that specialise in high-technology segments and make more intensive use of high-skilled labour. The lowest AI exposure values are predominantly found in South-Eastern European countries, including Italy, Spain and Romania. Greece and Portugal stand out with relatively higher AI exposure[29]. This pattern of relative

[29] Relatively higher AI exposure in Greece and Portugal can be attributed to several occupations with significant employment shares and high AI exposure scores, reflecting broader trends in their sectoral specialisations. More specifically, shop salespersons exhibit considerable employment shares in Greece (9.17%) and Portugal (5.4%); general office clerks also contribute significantly, with employment shares of 5.48% in Greece and 5.1% in Portugal, well above the EU average of 2.9%, along with higher shares of secondary education teachers.

AI exposure coincides neatly with other relevant dimensions, including skill levels, sectoral specialisation and technological capabilities across countries.

Focusing on the most exposed occupations (those in the top quartile of the AI exposure indicator), Figure 6 ranks EU Member States according to their share of total employment. It also shows the distribution of predominant task types within these professions.

Figure 6 Top AI-exposed occupations, by predominant task type, 2021



Notes: Calculates share of NRCA, NRCP and RC occupations in total employment for each country for occupations with the highest AI exposure (top quartile).
Source: Authors' elaboration.

Countries with a comparatively larger share of occupations dominated by NRCA tasks (e.g. mathematicians, finance and legal professionals), which appear to have been complementary so far, might gain most from AI adoption through enhanced worker productivity and, in turn, increased labour demand. Luxembourg, Finland, and the Netherlands exemplify this scenario. Conversely, countries with a higher share of occupations dominated by RC tasks (e.g. numerical and general office clerks, secretaries) might face more challenges as AI adoption accelerates. Italy, Greece, and Spain are cases in point, as the only Member States whose RC share within the highly AI-exposed segment surpasses that of NRCA occupations. The share of occupations most exposed to AI places Germany as a median case within the EU context, reflecting its diverse economic structure, including a strong manufacturing base. By contrast, Eastern European Member States exhibit significantly lower AI exposure. This difference is likely attributable to their specialisation in fabrication activities, unlike Germany or Austria, which specialise in other segments of the value chain, such as R&D (Kordalska et al., 2022). These regional differences in AI exposure and adoption reflect well-known structural differences, which will be further explored in the econometric analysis.

AI-EMPLOYMENT NEXUS: REGRESSION ANALYSIS

This section examines the relationship between employment growth rates and AI exposure at country-occupational level using regression analysis. Equations 3 and 4 are estimated to quantify the impact of AI exposure on annual employment growth rate in country i and occupation j over the period 2012-2022. The results are presented in Table 4.

Firstly, the average effect of AI exposure on employment growth is analysed (Table 4, columns 1-3). To test whether the AI-employment relationship varies across occupational groups and, more specifically, according to the relative share of RC tasks characterising each occupation, an interaction term is included, as specified in Equation 4, i.e. the AI exposure indicator is interacted with a binary variable equals to 1 for RC occupations and 0 for others (Table 4, columns 4-5).

The AI exposure coefficients are consistently positive and statistically significant across all specifications, indicating a positive association between AI exposure and employment growth in the EU. However, the magnitude of these coefficients diminishes as more control variables and country fixed-effects are introduced. When workforce characteristics such as the shares of native citizens, senior workers, men and permanent contracts, and educational attainment levels (upper secondary education and university degrees) are controlled for, the size of the coefficients is reduced (Table 4, columns 2-3). For instance, specification 3 suggests that, all else being equal, a 1 standard deviation increase in AI exposure is associated with an increase in annual employment growth of 0.46 percentage points (pp).

When the model is augmented to account for the interaction between AI exposure and task characteristics (RC tasks), the coefficient for AI exposure remains positive and significant, while the interaction term is negative. According to the RBTC hypothesis, this negative sign suggests that AI may not benefit RC-based jobs as much as non-RC-based jobs. However, as the interaction term is not statistically significant, it does not appear that AI has a labour-saving effect in occupations dominated by RC tasks.

Table 4. AI exposure and employment growth rate, 2012-2022

	(1)	(2)	(3)	(4)	(5)
AI exposure	1.359***	0.420*	0.463*	0.703**	0.646**
	(0.193)	(0.237)	(0.266)	(0.271)	(0.251)
AI exposure x RC				-0.552	-0.381
				(0.584)	(0.609)
RC tasks				-0.812*	-0.527

				(0.458)	(0.423)
% native		0.153	1.226	1.413	0.0990
		(1.228)	(1.837)	(1.839)	(1.138)
% 50-64		-7.221***	-6.833***	-6.836***	-6.843***
		(1.065)	(1.100)	(1.092)	(1.110)
% men		0.586	0.493	0.0808	0.490
		(0.420)	(0.438)	(0.494)	(0.419)
% upper secondary		-1.656*	-1.844*	-1.646	-1.576
		(0.834)	(1.008)	(1.013)	(0.996)
% tertiary		2.443**	2.027*	1.538	2.003*
		(0.947)	(1.103)	(1.092)	(0.996)
% permanent		-0.938	-0.796	-0.676	-1.056
		(0.657)	(0.765)	(0.764)	(0.777)
Public R&D (%GDP)					0.912
					(2.597)
Public procurement (%GDP)					0.00188
					(0.0730)
GVC gains (F/B)					-0.693
					(1.245)
Country FE	No	No	Yes	Yes	No
Constant	1.994***	4.068***	2.638	2.728	4.604**
	(0.265)	(1.253)	(1.648)	(1.632)	(2.145)
Observations	2,449	2,298	2,298	2,298	2,227

*Notes: Robust standard errors clustered at country level in parentheses *** p<0.01, ** p<0.05, * p<0.1; dependent variable is total employment growth over 2012-2022; all explanatory variables refer to 2012.*
Source: Authors' elaborations.

Cross-country heterogeneity is explored by splitting the sample into four country groups based on their innovation performance[30]: *Innovation Leaders*, *Strong Innovators*, *Moderate Innovators* and *Emerging Innovators* (European Commission, 2023). This classification reflects distinct national innovation systems and structural features that may affect the AI-labour nexus.

While the positive employment effect of AI exposure prevails in the EU, these gains are asymmetrically distributed. Table 5 shows that AI has a positive impact on employment growth only among *Innovation Leaders* (Belgium, Denmark, Finland, the Netherlands, Sweden) and *Strong Innovators* (Austria, Cyprus, France, Germany, Ireland, Luxembourg). Conversely, the magnitude of the coefficient

[30] This country grouping largely overlaps with the AI exposure distribution reported in Figure 5 and with varying AI adoption rates in the EU: Northern and Western EU countries lead in AI exposure/adoption, while Southern and Eastern countries lag behind (https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Use_of_artificial_intelligence_in_enterprises).

is markedly lower and statistically insignificant for *Moderate Innovators (*Czechia, Estonia, Greece, Hungary, Italy, Lithuania, Portugal, Spain) and *Emerging Innovators* (Croatia, Latvia, Poland, Romania, Slovakia).

These results suggest that the impact of AI is highly context-dependent: a country's innovation capabilities significantly influence how AI technologies can be harnessed to drive employment (and economic) growth. The positive employment effects are concentrated in countries with stronger research and innovation systems, characterised by higher R&D expenditure, more patent applications, a higher share of innovative enterprises, greater knowledge-intensive activities, and more attractive research environments. By contrast, the potential benefits of AI technologies are less likely to be realised in countries with moderate or emerging innovation systems.

Consistent with earlier findings (Albanesi et al., 2023; Guarascio at al., 2023), the positive employment effects of AI exposure in the EU (Table 4) should be interpreted with caution, as AI adoption and diffusion, and its implications for labour markets, differ significantly from one location to the other (Table 5).

Table 5. AI exposure and employment growth rate, by innovation group, 2012-2022

	(1)	(2)	(3)	(4)
	DK, SE, FI, NL, BE Innovation Leaders	AT, DE, LU, IE, CY, FR Strong Innovators	EE, CZ, IT, ES, PT, LT, EL, HU Moderate Innovators	HR, SK, PL, LV, RO Emerging Innovators
AI exposure	0.853**	1.192**	0.143	0.651
	(0.285)	(0.351)	(0.666)	(0.391)
AI exposure x RC	-0.520	0.279	-0.345	-1.627
	(0.761)	(1.282)	(0.719)	(2.079)
RC	-0.911	-2.182	-0.891*	0.990
	(0.468)	(1.128)	(0.450)	(1.287)
% native	8.367	2.218	-1.376	-0.798
	(5.138)	(2.664)	(3.336)	(1.681)
% 50-64	-6.634*	-10.24***	-6.840***	-4.190*
	(2.950)	(1.470)	(1.763)	(1.835)
% men	-0.833	0.273	0.403	0.0194
	(0.531)	(1.756)	(0.739)	(0.683)
% upper secondary	-2.863	-4.101	0.405	-2.282
	(2.219)	(2.859)	(1.467)	(1.457)
% tertiary	-0.713	0.524	4.054	0.652
	(1.490)	(1.229)	(2.232)	(1.555)
% permanent	-0.322	1.330	-1.860	-0.176
	(1.889)	(2.191)	(1.313)	(0.906)
Country FE	Yes	Yes	Yes	Yes

Constant	-2.643	2.544	4.653	5.334
	(4.279)	(2.157)	(3.792)	(2.641)
Observations	518	552	783	445
R-squared	0.134	0.169	0.106	0.100

*Notes: Robust standard errors clustered at country level in parentheses *** p<0.01, ** p<0.05, * p<0.1; dependent variable is total employment growth over 2012-2022; all explanatory variables refer to 2012. Source: Authors' elaborations.*

DISCUSSION AND POLICY IMPLICATIONS

The employment impact of AI is at the centre of a lively academic and policy debate, given its capacity to replicate – and in some cases exceed – an increasing number of human abilities in performing a wide array of tasks (Elondou et al., 2023). Views diverge into two camps: those optimistic about AI complementing labour and favouring the emergence of new sectors and markets, and those concerned about AI substituting workers and causing technological unemployment (Acemoglu, 2024; Autor, 2022). Current evidence is not conclusive, but tends to support the optimistic view. AI diffusion is still in its early stages, making its broader implications difficult to predict precisely and comprehensively.

This research note examines AI exposure and its employment impact within the EU, relying on state-of-the-art indicators (Felten et al., 2021; Acemoglu et al., 2022) for an in-depth analysis at occupational and country level. These indicators are useful for forming a preliminary understanding of the potential effects and structural changes stemming from AI in the near future.

The occupation-level analysis reveals that the highest levels of AI exposure are typically found among professionals (ISCO 2), a group characterised by higher levels of education and a prevalence of quantitative and analytical skills. More specifically, the most AI-exposed are mathematicians, actuaries, statisticians, and finance and legal professionals. Conversely, occupations least exposed to AI are generally manual and characterised by lower skill levels, notably elementary occupations (ISCO 9). This pattern is consistent across all sector where occupations predominantly involving physical tasks remain relatively unexposed to AI (agriculture, mining, manufacturing and services). These findings corroborate earlier studies (Felten et al., 2021; Albanesi et al., 2023; Elondou et al., 2023; Cazzaniga et al., 2024).

As higher exposure does not necessarily indicate a higher risk of substitution, it is useful to assess the nature of tasks performed within top AI-exposed occupations (Autor et al., 2003; Acemoglu and

Autor, 2011). The analysis shows that AI exposure is positively associated with employment growth, particularly in occupations dominated by NRCA tasks. This suggests that AI technologies are more likely to complement rather than substitute human labour in these roles, enhancing individual productivity, at least for now.

AI exposure and the nature of the tasks performed varies considerably across socio-demographic groups. In line with Cazzaniga et al. (2024) and Gmyrek et al. (2023), this analysis reveals a potentially gendered impact of AI: women are overrepresented in the most AI-exposed occupations (54.3%), compared to their overall presence in the workforce (46.8%). This disparity is even more pronounced in RC occupations, where women make up 75.7% of employees. In highly AI-exposed occupations dominated by non-RC tasks, 77% of workers hold a tertiary education degree, compared to only 39.2% in RC roles. This gender and skill disparity in highly AI-exposed occupations underscores the need for careful policy consideration to avoid exacerbating existing labour market inequalities and ensure that benefits of AI diffusion are more equitably distributed.

The regression results document a positive and significant association between AI exposure and employment growth, consistently with earlier studies (Albanesi et al., 2023; Damioli et al., 2023; Felten et al., 2019). However, the impact of AI varies across countries, benefiting only those with robust innovation systems such as *Innovation Leaders* (Belgium, Denmark, Finland, the Netherlands, Sweden) and *Strong Innovators* (Austria, Cyprus, France, Germany, Ireland, Luxembourg). By contrast, employment gains from AI are not evident in *Moderate Innovators (*Czechia, Estonia, Greece, Hungary, Italy, Lithuania, Portugal, Spain) and *Emerging Innovators* (Croatia, Latvia, Poland, Romania, Slovakia). This territorial disparity highlights the asymmetric distribution of innovation capability and the importance of their absorptive capacity to fully leverage the potential of AI. These preliminary findings require further micro-level empirical analysis; nevertheless, the insights suggest several policy implications.

To achieve broader and more equitable gains from AI across the EU, policies tailored to the specific needs of each country are likely to be more effective. In countries with emerging or moderate innovation systems, national innovation capability plays an important role in harnessing the benefits of AI. This typically involves high R&D expenditure and strong product innovation across firms. In less technologically advanced regions, strategic EU industrial and innovation policies (e.g. creation of innovation hubs in less developed areas) could be implemented to foster convergence at EU level.

While the overall effect of AI on employment appears positive, parallel labour-saving dynamics affecting specific occupations and sociodemographic groups remain a possibility (Acemoglu, 2024). The educational divide in occupations likely to benefit from AI highlights the need for substantial investment in upskilling and reskilling programmes. These programmes are designed to facilitate smooth transitions to other employers or sectors and to promote internal functional flexibility, enabling employees to shift to new positions within the same firm. Public employment services (PES) play a pivotal role in facilitating these transitions by providing comprehensive support, job search assistance and placement services (Ernst et al., 2024).

Active labour market policies (ALMPs) alone are unlikely to be sufficient to address the potential challenges posed by AI. To avoid a polarisation between economically and technologically stronger areas that are better placed to reap the benefits of AI, and weaker areas where such technologies are more likely to be imported and employed with labour-saving purposes, ALMPs could be accompanied by targeted macroeconomic and industrial policy initiatives capable of reducing these imbalances. As reskilling and upskilling are lengthy and complex processes rather than quick fixes, labour market policies need to be finely tuned to ensure that AI-related job opportunities match the existing skills of those facing the highest substitution risks (Minsky, 1965; Mazzucato, 2013; Rodrik and Stancheva, 2021; Rodrik, 2022).

Although this research note focuses on the employment impact of AI, its potential effects on job quality and income distribution must also be acknowledged (De Stefano and Doellgast, 2023). Two elements are worth underlining. Firstly, the development of AI technologies increasingly depends on platform workers who perform crucial yet underpaid micro-tasks, such as labelling sounds and images, transcribing audio, and correcting AI errors (Casilli, 2020). As the diffusion of AI applications accelerates, the share of precarious low-paid digital work is thus likely to increase. Secondly, AI is reshaping managerial practices in ways that may erode job quality and weaken workers' bargaining power: when managerial decisions are AI-based, they are harder to question, potentially substantially resizing the room for bargaining. In addition, the intensification of workplace surveillance and acceleration of work rhythms that are often associated with algorithmic management (Prassl, 2018) can deteriorate the work environment, heightening workers' social isolation and cognitive stress (Milanez, 2023; Staccioli and Virgillito, 2024).

The complementarity between AI and high-skilled jobs may lead to disproportionately higher wages for certain occupations and sectors, deepening labour income inequalities by benefiting those already well-positioned within the labour market (Cazzaniga et al., 2024). Policy plays an important role in ensuring that AI technologies are developed and deployed in ways that do not exacerbate labour market fragmentation or weaken workers' bargaining power (Prassl, 2018). Effective regulation of AI in workplace management and hiring practices is essential to mitigate these risks and promote fair and inclusive labour markets.

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APPENDIX

Table A1. AI exposure scores, by country, 2021

Country code	AICE
LU	3.02
SE	1.25
NL	1.24
MT	1.10
BE	0.83
IE	0.69
DK	0.49
FR	0.28
EL	0.26
FI	0.26
PT	0.22
DE	0.17
SI	0.15
AT	-0.04
CY	-0.13
LT	-0.14
EE	-0.21
PL	-0.27
LV	-0.52
CZ	-0.62
HR	-0.89
SK	-0.89
ES	-0.90
HU	-0.98
IT	-1.01
BG	-1.56
RO	-1.79

Source: Authors' elaboration.